Introduction to Large Language Models
Assignment- 8

TYPE OF QUESTION: MCQ
Number of questions: 10 **Total mark: 10 X 1 = 10**

QUESTION 1: [1 mark]

Which of the following statements about instruction tuning of a pretrained language model are true?

a) The prompt tokens are masked out of the next-token loss, so only the response tokens are supervised.
b) It updates the parameters of the base model.
c) It requires a separately trained reward model.
d) Training on several different phrasings of the same task improves generalisation to unseen instructions.

Correct Answers: a, b, d

Explanation:

(a) True: in a decoder-only model the instruction and response are concatenated, but cross-entropy is computed only over the response tokens. The prompt serves as conditioning context; the model is not being trained to predict the instruction itself.
(b) True: instruction tuning is supervised fine-tuning, so the weights of the base model are updated. This distinguishes it from pure prompting.
(c) False: a reward model belongs to the preference-optimisation stage (RLHF). Instruction tuning needs only (instruction, response) pairs.
(d) True: multiple templates expose the model to varied phrasings of the same underlying task, so it learns to follow instructions in general rather than memorising a few fixed formats.

QUESTION 2: [1 mark]

In prompt tuning, which parameters are updated during training?

a) All the weights of the pretrained model
b) Only the continuous prompt embeddings prepended to the input, while the model stays frozen
c) Only the final layer normalisation and the output head
d) The merge rules of the tokenizer

Correct Answer: b

Explanation:

Prompt tuning learns a small set of soft prompt vectors that are prepended to the input embeddings, and back-propagates only into those vectors. The backbone is frozen, so one copy of the model can serve many tasks by swapping in a different learned prompt.
(a) describes full fine-tuning. (c) describes a different parameter-efficient variant. (d) is unrelated - the tokenizer is fixed.

QUESTION 3: [1 mark]

Which capability does Tree-of-Thought (ToT) prompting add over Chain-of-Thought (CoT) prompting?

a) It fine-tunes the model on reasoning traces before inference.

b) It maintains several candidate reasoning branches and can backtrack from unpromising ones.
c) It removes the need for any prompt design.
d) It guarantees that the final answer is factually correct.

Correct Answer: b

Explanation:
CoT produces a single linear reasoning chain; if an early step is wrong, the whole chain is wrong. ToT frames reasoning as search over a tree of partial thoughts, evaluating branches and abandoning weak ones, which recovers from bad intermediate steps.
(a) is false - ToT is an inference-time procedure with no weight updates. (c) is false: ToT still needs a thought generator and an evaluator prompt. (d) is false - no prompting strategy guarantees correctness.

QUESTION 4: [1 mark]

Which of the following is NOT captured by the POSIX metric for prompt sensitivity?

a) Diversity in the responses produced under intent-preserving prompt variations
b) Entropy of the distribution of response frequencies
c) The wall-clock time taken to generate a response
d) Variance in the log-likelihood assigned to the same response across prompt variations

Correct Answer: c

Explanation:
POSIX quantifies how much a model's output distribution shifts when a prompt is rephrased without changing its intent. It is built from response diversity, response entropy and the relative change in log-likelihood of the same response under different intent-aligned prompts.
Inference latency is a systems property and has no bearing on how sensitive the model's output distribution is to prompt phrasing, so (c) is not part of POSIX.

QUESTION 5: [1 mark]

Which conclusion about prompt sensitivity is supported by POSIX measurements across model families?

a) Prompt sensitivity decreases monotonically with parameter count.
b) A larger model within the same family can still be more prompt sensitive than a smaller one.
c) Prompt sensitivity is zero for any instruction-tuned model.
d) All models of the same parameter count have identical prompt sensitivity.

Correct Answer: b

Explanation:
The reported results show, for example, that a 13B model in a family is not guaranteed to be less prompt sensitive than the 7B model from the same family. Scale alone therefore does not resolve prompt sensitivity, which is why it must be measured separately from accuracy.
(c) is false - instruction tuning reduces but does not eliminate sensitivity. (d) is false, since sensitivity also depends on the training data and the tuning recipe.

QUESTION 6: [1 mark]

Why is alignment performed on top of instruction tuning rather than stopping at instruction tuning?

a) Instruction tuning does not allow the model to follow any instructions at all.
b) Instruction tuning can still yield fluent, instruction-following responses that humans judge harmful or unacceptable.

c) Instruction tuning destroys the model's performance on downstream tasks.

d) Alignment is needed only for models below a few billion parameters.

Correct Answer: b

Explanation:

Instruction tuning teaches the model what form a good answer takes, but the supervised targets do not encode human judgements about which of several plausible answers is preferable or safe. A model can obey the instruction literally and still produce a harmful response.

Alignment methods such as RLHF add exactly that missing signal by learning from human preference comparisons.

QUESTION 7: [1 mark]

A reward model r(x, y) is trained on preference triples (x, y_w, y_l), where y_w is the preferred response and y_l the rejected one. Using the Bradley-Terry formulation, the training objective maximises:

a) log sigma(r(x, y_l) - r(x, y_w))

b) log sigma(r(x, y_w) - r(x, y_l))

c) the entropy of the policy distribution

d) the KL divergence between the policy and the reference model

Correct Answer: b

Explanation:

The Bradley-Terry model expresses the probability that y_w is preferred over y_l as sigma(r(x, y_w) - r(x, y_l)). Maximum likelihood estimation on the observed human preferences therefore maximises the log of that sigmoid, which is option (b).

(a) has the sign reversed and would train the model to prefer the rejected response. (c) and (d) belong to the policy optimisation stage, not to reward model training.

QUESTION 8: [1 mark]

In RLHF the policy is optimised for reward while a KL penalty against a reference policy is applied. What is the primary purpose of that KL term?

a) To make each gradient update cheaper to compute

b) To keep the optimised policy close to the reference model, so it does not drift into degenerate high-reward behaviour

c) To increase the entropy of the sampled responses

d) To remove annotator bias from the preference dataset

Correct Answer: b

Explanation:

The learned reward model is only an approximation of human preference, so a policy left free to maximise it will find inputs where the proxy is wrong - reward hacking - and lose the fluency acquired during pretraining and instruction tuning. Penalising KL divergence from the reference policy bounds how far the distribution can move while still exploiting the reward signal.

(a) is false: the penalty adds computation rather than saving it. (c) is a side effect at best, not the purpose. (d) is false - a KL term cannot correct bias present in the preference data.

QUESTION 9: [1 mark]

Which of the following statements about REINFORCE-style policy optimisation for language models are true?

a) The log-derivative trick yields an unbiased estimator of the policy gradient from sampled responses.

b) Weighting the token-level log-probabilities by an advantage rather than the raw return reduces gradient variance.

c) Importance ratios are clipped when several updates are taken on samples drawn from an older policy.

d) The reward function must be differentiable with respect to the model parameters.

Correct Answers: a, b, c

Explanation:

(a) True: rewriting the gradient using grad log pi lets the gradient of an expectation be estimated by sampling, which is the foundation of policy gradient methods.

(b) True: the raw return has high variance. Subtracting a value baseline to form an advantage keeps the estimator unbiased while substantially reducing its variance.

(c) True: reusing samples from an old policy requires importance weighting, and clipping those ratios is what keeps the updates stable - the core idea of PPO.

(d) False: this is precisely why policy gradients are used. The reward may be a discrete or non-differentiable human or model score, and the log-derivative trick avoids differentiating it.

QUESTION 10: [1 mark]

Let ratio_t = pi_new(a_t | s_t) / pi_old(a_t | s_t), where pi_new is the policy being optimised and pi_old the policy the samples were drawn from, and let epsilon > 0 be the clipping parameter. When the token-level advantage A_t is negative, the PPO-CLIP objective at step t reduces to maximising:

a) min(ratio_t, 1 + epsilon) A_t

b) max(ratio_t, 1 - epsilon) A_t

c) min(ratio_t, 1 - epsilon) A_t

d) max(ratio_t, 1 + epsilon) A_t

Correct Answer: b

Explanation:

The PPO-CLIP objective is min(ratio_t * A_t , clip(ratio_t, 1 - epsilon, 1 + epsilon) * A_t).

With A_t negative, multiplying by A_t reverses the order, so the minimum over the two products corresponds to the maximum over the two ratios: A_t * max(ratio_t, clip(ratio_t, 1 - epsilon, 1 + epsilon)), which simplifies to A_t * max(ratio_t, 1 - epsilon).

Intuitively, a negative advantage means the action was worse than average and its probability should fall, but the clip stops the ratio from being pushed below 1 - epsilon in a single update. This is the mirror image of the positive-advantage case, where the objective is min(ratio_t, 1 + epsilon) A_t.
